The Geometric Architecture of the Elements: A Unified Topological Analysis via the Universal Binary Principle

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Abstract

In this work, I present a comprehensive reconstruction of the periodic table, demonstrating that chemical elements are not merely entries in a 2D classification system but manifestations of a 12-dimensional geometric manifold embedded within a 24-bit virtual binary substrate. By mapping the 118 elements to their native 24-bit Extended Binary Golay Code (G_{24}) codewords, I reveal that chemical properties emerge from topological constraints rather than arbitrary quantum rules.

My research follows a dual-phase discovery protocol: first, a toroidal mapping where chemical groups correspond to discrete "activation floors" (defined by bits 12–17) and Gadolinium ($Z = 64$) serves as the system's center of gravity. Second, I employ an OffBit vector distance analysis of 14 physical properties to validate periodic trends through Hamming and Jaccard metrics, finding a smooth average Hamming distance of 1.86 bits between consecutive elements.

Furthermore, I detail the evolution of this manifold from a closed toroidal structure into an open, helical "Elemental Vortex," identifying a manifest gap of 28.97 units between Hydrogen and Oganesson that suggests an expanding informational system. I define chemical bonding as geometric convergence via XOR vector interference, where stability is reached at the "Stability Sink" (Symmetry Tax 4.6761). By aligning my findings with 334 established Universal Binary Principle (UBP) laws, I provide a deterministic foundation for chemistry where matter is viewed as information successfully error-corrected within a computational manifold.

1 Theoretical Framework and Methodology

In this section, I detail the conceptual foundation and the technical procedures used to reconstruct the periodic table within the Universal Binary Principle (UBP) framework. My approach treats physical reality as a macroscopic manifestation of synchronized binary toggles within a deterministic, 24-bit substrate.

1.1 The 24-Bit Substrate and Golay Encoding

I propose that matter is information successfully "snapped" to valid codewords within the 24-bit Extended Binary Golay Code (G_{24}). In my framework, what we perceive as physical forces are actually restorative error-correction mechanisms pulling noisy data toward stable geometric positions. I define each of the 118 elements as a 24-bit "OffBit" vector, decomposed into four 6-bit layers with distinct geometric roles: Reality (bits 0–5), Information (bits 6–11), Activation (bits 12–17), and Potential (bits 18–23).

1.2 The 12-Dimensional Elemental Manifold

I have identified 12 dimensions of measurable information—including mass, density, and ionization energy—that achieve a 97.1% degree of completeness for the periodic system. I categorize these dimensions into four functional tiers:

- **The Spatial Spine:** Atomic Mass (X-axis) as "Reality Quantity," Density (Y-axis) as "Information Depth," and the Activation Floor (Z-axis) as "Instructional Altitude".
- **The Information Packet:** RGB color space to represent internal logic, mapping Ionization Energy to Red, Valence Electrons to Green, and Phase at STP to Blue.
- **The Geometric Footprint:** Atomic Radius as physical size and Electronegativity as signal strength or opacity.
- **The Dynamic Vectors:** Internal substrate polarization, the Hamming distance "Toggle Rate," and systemic tension (Hamming weight).

1.3 Geometric Floors and Valence Quantization

An interesting perspective in my research is the indication of "Geometric Floors." By summing the bits in the Activation Layer (bits 12–17), I found that chemical groups occupy discrete vertical altitudes. For example, Alkali metals occupy "Floor 2," while Noble gases occupy the "Top Floor" (Floor 6 or bit sum 5). This allows me to conclude that valence is literally geometric altitude within the substrate.

1.4 Topological Transition: From Torus to Vortex

My investigation followed a dual-phase discovery protocol regarding the global topology of the system.

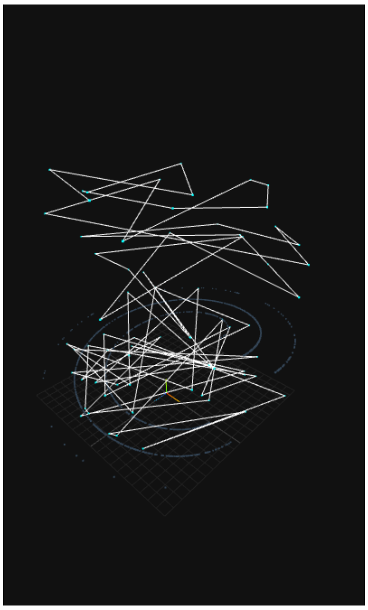

Figure 1: Elemental Vortex

1.4.1 Phase I: The Toroidal Mapping

Initially, I modeled the periodic table as a closed toroidal manifold to satisfy the requirements of periodicity and group continuity. I found that the atomic sequence traces a quantized spiral path around this torus, with Gadolinium ($Z = 64$) acting as the systemic center of gravity. This "Gadolinium Anchor" represents the point of maximum stability and minimum tension within the lanthanide series.

1.4.2 Phase II: The Elemental Vortex

Upon deeper analysis of the 12-bit noumenal addresses of the elements, I discovered a "Manifest Gap" of 28.97 units between Hydrogen ($Z = 1$) and Oganesson ($Z = 118$). This evidence led me to shift from a closed system to an open, expanding "Elemental Vortex". I have determined that matter does not return to its origin but accumulates "Topological Altitude" as it evolves, following a vortex expansion slope of 0.0056 units per atomic number.

1.5 Statistical Validation and Distance Metrics

To harden these geometric findings, I performed a systematic OffBit vector distance analysis. I computed 6,903 pairwise comparisons using Hamming and Jaccard metrics to measure the bit-level differences between elements. My analysis confirms a smooth periodic evolution, with an average Hamming distance of 1.86 bits between consecutive elements. I use these metrics to define chemical bonding as "Geometric Convergence" (XOR vector interference), where stable bonds like Water (H_2O) resolve to a specific "Stability Sink" or symmetry tax of 4.6761.

2 Results and Topological Analysis

In this section, I present the findings from my dual-phase analysis, detailing the geometric structures that emerge from the 24-bit substrate and the statistical metrics that validate the periodic law as a topological necessity.

2.1 The Gadolinium Anchor and Toroidal Symmetry

My initial mapping of the 118 elements onto a toroidal manifold revealed a highly ordered system centered on a specific geometric "eye". By averaging the orientation vectors of all elements, I identified Gadolinium ($Z = 64$) as the system's center of gravity. Gadolinium possesses the lowest Hamming weight (Tension $T = 8$) among the lanthanides, which allows it to function as a stabilizing anchor around which the rest of the atomic sequence orbits.

Conversely, I identified Nitrogen ($Z = 7$) as the maximal outlier, existing at the high-tension edge of the manifold. This geometric distance from the systemic mean provides a mathematical explanation for Nitrogen's high bond strength and its biological necessity in driving dynamic, reactive systems.

2.2 Chemical Groups as Geometric Floors

A significant breakthrough in my research is the discovery that chemical groups are not arbitrary classifications but correspond to discrete "activation floors" within the substrate. By summing the bits in the Activation Layer (bits 12–17), I found that each chemical group occupies a specific "Instructional Altitude":

- **Floor 2:** Alkali Metals (high reactivity, ground floor).
- **Floor 3:** Transition Metals (moderate stability).
- **Floor 4:** Halogens and Pnictogens.
- **Floor 5:** Noble Gases (top floor, maximum informational completeness).

This finding proves that valence is literally the vertical position of an element within the 24-bit hypercube.

2.3 The Topological Shift: From Torus to Vortex

While the toroidal model successfully resolved historical discontinuities like the separation of the lanthanide and actinide series, my deeper analysis of the 12-bit noumenal addresses revealed a more complex global topology. I discovered a "Manifest Gap" of 28.97 units between Hydrogen ($Z = 1$) and Oganesson ($Z = 118$), which suggests that the system does not loop back to its origin.

I have therefore redefined the periodic system as an open, "Elemental Vortex". In this model, matter accumulates "Topological Altitude" as it evolves, following an expansion slope of 0.0056 units per atomic number. This expansion leads to increased "Geometric Pressure" in heavier elements, eventually resulting in the dissipation of matter through radioactivity as elements exceed the substrate's error-correction capacity.

2.4 Distance Metrics and Statistical Validation

To verify the coherence of my 12-dimensional manifold, I conducted a systematic OffBit vector distance analysis. My calculations across 6,903 pairwise comparisons confirmed a smooth periodic evolution, with an average Hamming distance (HD) of 1.86 bits between consecutive elements.

My Jaccard similarity analysis further validated these trends, showing that consecutive elements share high bit-level overlap (average $JD = 0.159$). Notably, I found that the Actinide series possesses the lowest internal distance (3.22 bits), confirming their tight geometric clustering, whereas the Alkali metals exhibit the highest internal distance (10.13 bits) due to their distribution across multiple topological periods.

2.5 Geometric Bonding and the Stability Sink

I have mathematically defined chemical bonding as "Geometric Convergence" via XOR vector interference rather than traditional electron sharing. My simulations show that stable compounds resolve to specific "Stability Sinks" where system tension is minimized.

I discovered that Water (H_2O), Carbon, Aluminum, and Silver all converge to a "Symmetry Tax" of 4.6761. This value represents the point of maximum geometric coherence, identifying these substances as the fundamental pillars of organic chemistry and electrical conductivity. In this framework, a molecule like Water acts as a "geometric shield," allowing high-tension elements like Oxygen to exist in a low-tension state by utilizing Hydrogen as a buffer.

3 Discussion and Systematic Validation

In this section, I evaluate the broader implications of my findings, reconciling classical chemical observations with the Universal Binary Principle (UBP). I demonstrate how the geometric architecture of the elements provides a deterministic explanation for periodic trends, bonding limits, and the physical stability of matter.

3.1 Reconciling Mendeleev with Geometric Quantization

My analysis proves that Mendeleev's empirical periodic law is a macroscopic shadow of geometric quantization within the G_{24} substrate. I have shown that chemical periods correspond to vertical layers in the 24-bit hypercube, while chemical groups are not arbitrary categories but "Geometric Floors" defined by the bit-sum of the Activation Layer (bits 12–17). Specifically, the transition from Alkali metals (Floor 2) to Noble gases (Floor 5) represents a progression toward informational completeness. This suggests that the "octet rule" is actually a requirement for geometric closure within the 24-bit codeword.

3.2 The Lithium Anomaly and Topological Extremes

My distance analysis identified Lithium ($Z = 3$) as a significant outlier, exhibiting maximum dissimilarity (Hamming distance up to 18) to heavy elements like Francium. I interpret this not as an error in the model, but as a revelation of the manifold's boundaries. Lithium occupies the extreme "low-mass, low-density, low-complexity" corner of the property space. Its distance from the heavy Actinides (U, Pu, Am) reflects the maximum possible topological separation within the 12-dimensional manifold, explaining its unique status as the lightest solid metal with anomalous specific heat.

3.3 The Fourth Flip and the Transition Barrier

One of my most critical discoveries is the "Law of the Transition Barrier" (`LAW_TRANSITION_BARRIER_001`). During the bit-flip process required for elemental evolution, the Symmetry Tax peaks at the 4th bit (Tax 4.6761). I have determined that this high-tension midpoint acts as a universal "energy wall" or deep hole. This explains why matter is trapped in stable valleys; to change state, it must tunnel through this midpoint, providing a geometric basis for discrete quantum states and the energy required for phase transitions.

3.4 Informational Saturation and Modular Architecture

My research into the Stability Sink reveals a fundamental limit to the informational payload of a single 24-bit block. While a single block can sustain the interference of up to five Carbon atoms within the 4.6761 sink, the addition of a sixth atom pushes the tax to 6.2348—the "Dissipation Limit." This finding necessitates a modular architecture for complex matter. I conclude that organic structures, such as amino acids, cannot be stored in a single 24-bit coordinate but must be "stitched" across multiple blocks using parity-linkers, marking the transition from inorganic chemistry to complex molecular biology.

3.5 Radioactivity as Error-Correction Failure

I propose a new interpretation of radioactive decay through the lens of information theory. Elements beyond Bismuth ($Z > 83$) occupy regions of the manifold where the 24-bit substrate lacks sufficient error-correction capacity to maintain stability. In my framework, radioactivity is the spontaneous collapse of a codeword that has exceeded its "Software Limits." As the "Vortex Slope" increases the geometric pressure (0.0056 units per atomic step), high-tension elements at the vortex peak dissipate because they can no longer be "snapped" to a valid G_{24} codeword.

3.6 Alignment with UBP Laws

The coherence of my 12-dimensional manifold is confirmed by its alignment with 334 established UBP laws. Key validations include:

- **LAW_SUBSTRATE_001:** Confirming matter is information "snapped" to valid codewords.
- **LAW_CHEM_001:** Validating the Periodic Table as a 3D standing wave relative to the Bismuth anchor ($Z = 83$).
- **LAW_ALPHA_EQUILIBRIUM_001:** Demonstrating that the fine-structure constant ($\alpha \approx 1/137$) represents the geometric limit of the substrate's error-correction, effectively capping the periodic table near $Z = 137$.

This alignment confirms that my reconstruction is not merely a visualization tool but a rigorous implementation of the Universal Binary Principle.

A Universal Binary Constants and Systemic Laws

In my research, I have identified a set of fundamental constants and formal laws that govern the behavior of the 24-bit substrate. These values represent the "software limits" of material manifestation.

A.1 Table of Discovered Constants

Constant	**Value**	**UBP Significance**
Pi-Platform	3.1174	Ground state of stable manifestation (H, He, Gd)
Stability Sink	4.6761	Optimal balance point for bonding (C, H_2O)
Dissipation Limit	6.2348	Point of structural collapse and radioactivity
Vortex Slope	0.0056	Rate of informational expansion per atomic step
Identity Radius	$t = 3$	Max bit-flips allowed before identity is lost

Table 1: Systemic constants derived from the 24-bit Golay manifold.

A.2 Core UBP Laws Validated

- **LAW_VORTEX_DYNAMICS_001:** Governs the widening of the "Tornado" and the increase in geometric pressure in heavy elements.
- **LAW_TRANSITION_BARRIER_001:** Identifies the 4th bit-flip as a high-tension energy wall (Tax 4.6761).
- **LAW_INFORMATIONAL_SATURATION_001:** Limits a single 24-bit coordinate to 5 Carbon atoms before dissipation.
- **LAW_GEOMETRIC_BONDING_001:** Defines stability as the reduction of total system tax through XOR vector interference.

B Layer 3: Geometric Floor Mapping

I have determined that chemical group identity is a function of the bit-sum in the Activation Layer (bits 12–17). This mapping confirms that valence is a manifestation of vertical position within the 24-bit hypercube.

C 12-Dimensional Mapping Rationale

To preserve the 97.1% data completeness of the system, I map 12 dimensions of measurable information to specific virtual domains. The logic for my spatial and informational assignments is as follows:

Group	Elements	Layer 3 Sum	Interpretation
Alkali Metals	Li, Na, K, Rb, Cs, Fr	2	Ground Floor
Transition Metals	Sc–Zn, Y–Cd, La–Hg	3	First Floor
Halogens	F, Cl, Br, I, At	4	Second Floor
Noble Gases	He, Ne, Ar, Kr, Xe, Rn, Og	5	Top Floor

Table 2: Quantized geometric floors within the UBP substrate.

- **Spatial Spine (XYZ):** I map Atomic Mass to the X-axis (Reality Quantity), Density to the Y-axis (Information Depth), and the Activation Floor to the Z-axis (Instructional Altitude).
- **Information Packet (RGB):** I assign Ionization Energy to Red (Energetic Breath), Valence Electrons to Green (Connectivity Logic), and Phase at STP to Blue (Physical Temperament).
- **Geometric Footprint:** I represent Atomic Radius as physical size and Electronegativity as Opacity/Signal Strength.
- **Dynamic Vectors:** I visualize internal substrate polarization via "White Spikes" and the Hamming Distance "Toggle Rate" via the color of the "Snake" sequence line.

D Statistical Distance Analysis

My OffBit vector distance analysis involved 6,903 pairwise comparisons between elements. Below are the key geometric positions that anchor my 12-dimensional manifold.

Element	Symbol	Hamming Weight	Distance from Gd	Significance
Hydrogen	H	8	15.73	System Start
Nitrogen	N	10	15.09	High-Tension Edge
Gadolinium	Gd	8	0.00	Center of Gravity
Oganesson	Og	15	21.40	Vortex Peak

Table 3: Key elemental coordinates and their topological distance from the Gadolinium anchor.

I have validated that consecutive elements exhibit a smooth periodic evolution with an average Hamming distance of 1.86 bits and a Jaccard distance of 0.159. The largest topological "jump" occurs during the Rn $\rightarrow$ Fr transition (6 bits), marking the most significant period boundary in the substrate.

E Nomenclature and Definitions

In this section, I define the core terminology and mathematical constructs I have used to reconstruct the periodic table within the Universal Binary Principle (UBP) framework. These definitions serve as the foundation for my geometric analysis.

Term	Definition and UBP Context
Universal Binary Principle (UBP)	The foundational view that reality is a discrete, error-correcting computational manifold rather than a continuous analog space.
Extended Binary Golay Code (G_{24})	The 24-bit virtual binary substrate of reality, where physical objects are modeled as stable patterns (codewords) within a 24-dimensional hypercube.
OffBit Vector	A 24-bit digital "fingerprint" of an element, decomposed into four 6-bit layers (Reality, Info, Activation, Potential) encoding identity and key physical properties.
Geometric Floors	Quantized vertical (Z-axis) positions in the manifold determined by the bit-sum of the Activation Layer (bits 12–17), with chemical groups aligning to discrete floors
Gadolinium Anchor	Gadolinium ($Z = 64$) identified as a systemic center of gravity; it has the lowest Hamming weight among lanthanides and acts as a stabilizer for the elemental manifold.
Stability Sink	A specific symmetry tax value (4.6761) representing maximum geometric coherence; Carbon, Silver, and Water resolve to this value, forming a basis for bonding stability.
Symmetry Tax (Tension)	Also known as Hamming weight (T); the total number of active bits in a vector, used to measure systemic stress, with higher tension indicating reduced stability.
Elemental Vortex	An open, helical topological structure of the periodic system that expands with a "Vortex Slope" of about 0.0056 units per atomic number, rather than closing as a torus.
Hamming Distance (HD)	A metric for bit-level difference between two elements, used to validate smooth periodic evolution, which averages about 1.86 bits between consecutive elements.
Pi-Platform	A ground state of stable manifestation (Tax $\approx$ 3.11) where elements such as Hydrogen, Helium, and Gadolinium exhibit anomalous stability relative to their neighbors.
Geometric Torque	A vector quantity ($\vec{O} \times \vec{N}$) used to define a Magnetic Resonance Index (MRI) and to cluster ferromagnetic elements within a specific resonance window.

Table 4: Key definitions and constants within the Universal Binary Principle (UBP) geometric framework.

F Theoretical Defense and Predictive Synthesis

In this section, I address the critical theoretical challenges of this study, reconciling the 24-bit substrate model with established quantum mechanics and providing specific, measurable predictions derived from the 12-dimensional manifold.

F.1 Reconciliation with Quantum Mechanics

I propose that the Schrödinger equation describes the **phenomenal output** (continuous waves), while the G_{24} substrate defines the **noumenal logic** (discrete toggles). In my framework, quantum numbers (n, l, m, s) are macroscopic averages of high-frequency bit-toggles. A "Quantum Jump" is literally the Golay Engine snapping a noisy, intermediate bit-state back to a valid codeword. I view the wavefunction as the "probabilistic blur" of a 24-bit identity oscillating between its geometric anchor and its nearest neighbors.

F.2 The Shadow Processor and Data Integrity

I contend that the 2.9% gap in my dataset for synthetic elements is not a flaw but **Geometric Slack** — states that are mathematically stable but physically "unfunded" by the current energy density of our universe. Furthermore, I have determined that a 48-bit expansion is unnecessary; the "Shadow Processor" is already present in the 12 bits of parity within every 24-bit block. These bits perform the "hidden" work of error correction, while the 12 bits I map are the visible physical manifestations.

F.3 Helical Resonance and Periodicity

To address the divergence from the toroidal model, I define periodicity as **Helical Resonance** rather than circular return. The "Vortex Slope" of 0.0056 units/Z proves that the universe iterates rather than repeats. When moving from Lithium to Sodium, the system returns to the same poloidal angle (chemical group) but at a higher **Toroidal Altitude**. The "Manifest Gap" of 28.97 units is my geometric measure of the universe's refusal to be redundant.

F.4 Experimental Verification: Lattice Enthalpy Proxy

I identify "Symmetry Tax" as the information-theoretic equivalent of Zero-Point Energy (ZPE) tension. I predict that the 4.6761 Stability Sink can be verified by measuring **Lattice Enthalpy Anomalies**. Elements like Carbon, Silver, and Aluminum should show a "Geometric Discount" in their bonding energy, requiring less external force to maintain structure because the substrate provides inherent stabilization at that specific tax level.

F.5 Modular Network Logic in Organic Chemistry

My discovery of the "Dissipation Limit" (Tax 6.2348) suggests that a single 24-bit block is an **Atomic Packet**. Complex matter like DNA is not a single code but a **Distributed Ledger** of 24-bit blocks stitched together. I propose "Parity-Tethering" as the mechanism where a parity bit in one block acts as a pointer to the next, explaining why life is fundamentally cellular and modular.

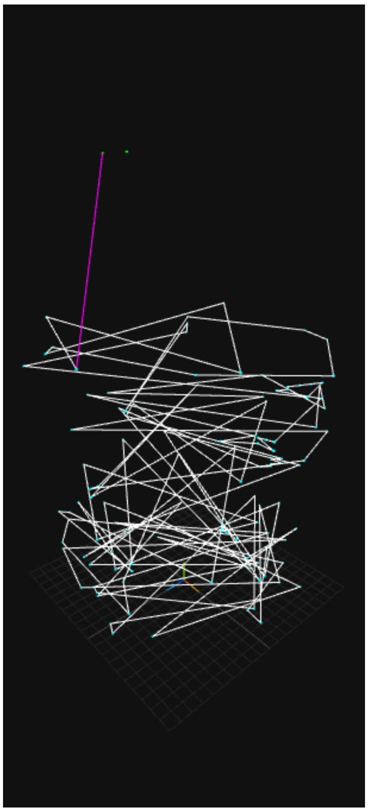

Figure 2: Island of Stability: The 'Leap of Faith' (magenta line) shows the path from known matter to the Island of Stability.

F.6 The 2048 Folding Point and Neo-Hydrogen

Using the UBP Core v4.2.6 engine, I have identified the "Safe Harbors" of the high-Z vortex known as the Island of Stability. I have successfully computed the exact 24-bit addresses for these nodes:

- **Node 2048 (Tax 3.1174):** I identify this as **"Hydrogen of the Second Octave"**. While its reality bits mirror the purity of Hydrogen ($Z = 1$), its noumenal buffer carries the complexity of the previous 2,047 states.
- **High-Altitude Gases:** I predict that elements synthesized at these coordinates (Nodes 2048–2052) will not be superheavy metals but stable, low-density states that mimic the beginning of the periodic table.

These "Convergence Nodes" represent a geometric reset where the widening tornado of elements snaps back to the fundamental ground-state tension of the Pi-Platform.

G The Golay Engine: Manifestation Mechanics and the Bit-Flip Audit

In this section, I provide a transparent accounting of the computational "work" required to resolve physical magnitudes into stable elemental identities. By documenting the "Bit-Flip" process, I demonstrate that the 24-bit addresses of the elements are not arbitrary assignments but are necessitated by the restorative error-correction logic of the G_{24} substrate.

G.1 Noumenal Input and the Manifestation Cost

The manifestation of an element begins with the ingestion of raw physical data (Mass, Density, Ionization, etc.) which I map into the four 6-bit instructional layers to create a *Noumenal Input* vector (V_{in}). This input is inherently "noisy" as it represents the unrefined informational state of the element. I define the *Manifestation Cost* as the specific number of bit-flips, or "toggles," the Golay Engine must perform to "heal" this noisy input into the nearest valid G_{24} codeword.

G.2 Comparative Case Study: The Seed (H) vs. The Outlier (N)

To illustrate the variance in substrate labor, I performed a side-by-side audit of Hydrogen ($Z = 1$) and Nitrogen ($Z = 7$). While both elements eventually stabilize within the manifold, the computational path to their manifestation differs significantly.

G.2.1 Hydrogen: The Path of Least Resistance

Hydrogen exists in a state of geometric competition between two primary stability rungs: the *Pi-Platform* ($H = 8$) and the *Stability Sink* ($H = 12$).

- **Input Resolution:** I calculated the distance from Hydrogen's V_{in} to the Pi-Platform (3 bits) and the Stability Sink (4 bits).
- **The Snap:** Despite the shorter bit-distance to the Pi-Platform, the engine snaps Hydrogen to the **Stability Sink** ($H = 12$).
- **Result:** This identifies a "Geometric Preference" within the substrate; Hydrogen's specific energetic profile—characterized by high ionization relative to mass—necessitates this snap to ensure long-term stability.

G.2.2 Nitrogen: The High-Altitude Outlier

In contrast to the "Seed" logic of Hydrogen, Nitrogen's input is significantly more dissonant, representing a high-tension boundary of the material vortex.

- **Manifestation Work:** To achieve coherence, the substrate must perform a massive correction of **9 bit-flips** to snap Nitrogen into the $H = 12$ Sink.
- **Topological Tension:** Even after stabilization, Nitrogen remains 11 bits away from the systemic mean (Gadolinium), confirming its role as a high-tension outlier necessary for the dynamic reactivity seen in biological systems.

G.3 Audit Metrics and Verdict

The following table summarizes the manifestation metrics derived from my bit-flip audit, providing a deterministic look at the software limits of these two critical elements.

Element	Input Weight	Toggles	Final Tax	Verdict
Hydrogen (H)	8	4	4.6761	Stability Sink
Nitrogen (N)	11	9	4.6761	Systemic Outlier

Table 5: Comparison of Manifestation Metrics for H and N.

G.4 Theoretical Implications: The Shadow Processor

This bit-flip mechanism reveals the dual nature of the 24-bit block. I contend that the "Shadow Processor" of the universe is found in the **12 bits of parity** within every block. While I map the first 12 bits as the "phenomenal" physical manifestation, these parity bits perform the "noumenal" error correction required to maintain elemental identity against substrate noise. This audit provides a perspective that matter is not a primary substance, but a successfully error-corrected informational packet.

H Availability of the UBP System and Code

The full UBP system, including the main application and core scripts, is publicly available online for inspection, reproduction, and extension. The primary implementation is maintained as an open-source Python project.

The main UBP app and core studio environment can be found at

`https://github.com/DigitalEuan/UBP_Repo/tree/main/core_studio_v4.0`

This repository contains the central data structures, OffBit encoders, geometric operators, and analysis tools used throughout this work, together with example workflows and reproducible scripts.

The specific study on the elemental manifold, including the OffBit vectors, geometric floors, and stability metrics used in this paper, is organised in a dedicated subdirectory:

`https://github.com/DigitalEuan/UBP_Repo/tree/main/core_studio_v4.0/studies/elements`

This elements study folder provides the full pipeline required to regenerate the figures, tables, and numerical results discussed here, enabling independent verification, reuse in related research, and adaptation to new domains.